MILITARY SPECIFICATION SHEET

ELECTRON TUBES, CATHODE RAY

TYPES 5AFP1 AND 5AFP7

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Dual beam, post-accelerator, flat face, electrostatic focus and deflection

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	Meg	Vdc	Ratio	ft
Maximum:	6.9	0, -200	1,750	3,500	10,500	750	1.5	1.0	±180	3.0	50,000
Minimum:	5.7	---	---	1,000	1,000	---	---	---	---	---	---
				(see note 1)				(see note 2)		(see note 3)	
TEST CONDITIONS: (see note 4)	6.3	Adjust	Focus	2,000	4,000	---	---	1.0	---	2.0	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection					
1216	Base material insulating quality		---	---	---	---
5101	Neck and bulb alinement (electrostatic types)		Dia	---	3.31	in.
5101	Face tilt		---	---	---	---
5216	Cathode illumination		---	---	---	---
1002	Barometric pressure, reduced	87 mmHg (see note 6)	---	---	---	---
1141	Pressure (implosion)		---	---	---	---
5111	Vibration		Width	---	1	mm
1331	Direct-interelectrode capacitance	See note 12				
		k to all	Ck	---	4.6	pF
		g1 to all	Cg1	---	4.0	pF
		D1 to D2	C1D2	---	2.0	pF
		D3 to D4	C3D4	---	1.4	pF
		D1 to all	CD1	---	5.0	pF
		D2 to all	CD2	---	5.0	pF
		D3 to all	CD3	---	4.1	pF
		D4 to all	CD4	---	4.1	pF
5248	Deflection-factor uniformity	1D2	---	---	2	%
		3D4	---	---	3	%
5250	Interaction factor		---	---	14×10^{-6}	In./Vdc
5256	Magnetization		---	---	---	---
	Quality conformance inspection, part 1					
5201	Voltage breakdown		---	---	---	---
5201	Voltage breakdown (electrostatic types)		---	---	---	---
5206	Gas "cross"	Ib3 = 50 μAdc (see note 5)	---	---	---	---
5223	Modulation	Ib3 = 25 μAdc	ΔEc1	---	35	Vdc
5231	Spot position (electrostatic deflection)		---	---	15	mm
5231	Spot displacement (leakage)		Displ	---	10	mm
5241	Grid cutoff voltage		Eco	-45	-75	Vdc
---	Tracking	See note 9	---	---	1.5	%
----	Useful scan	1D2 (see note 10)	---	---	4	in.
		3D4 (see note 10)	---	---	4	in.
5103	Pattern distortion	See note 11	---	---	2	%
5251	Grid No. 1 leakage current		---	---	---	---
5251	Anode No. 1 leakage current		Ib1	---	3	μAdc
5251	Anode No. 2 leakage current		---	---	---	---

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	METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
		Quality conformance inspection, part 1 -Continued					
Ⓑ	5221	Light output (P1)	Ib3 = 25 µAdc (see note 5)	Light	20	---	ftL
		Quality conformance inspection, part 2					
	1301	Heater current		If	540	660	mA
	5201	Electrode currents (anode No. 1)		Ib1	-10	5	µAdc
	5201	Electrode currents (cathode)	Ib3 = 50 µAdc	Ik	---	600	µAdc
	5101	Base alinement (electrostatic types)	+3D4, base key, collar index pin	---	---	---	---
	----	Collar base alinement	+3D4, collar index pin	---	---	10	Degrees
	5101	Side terminal and base alinement	Base key, collar index pin	---	---	---	---
	5101	Side terminal alinement (electrostatic types)	+3D4	---	---	---	---
	5101	Neck and base alinement (electrostatic types)		---	---	---	---
	5221	Screen (P7)	See note 5	---	---	---	---
	5226	Line width B (electrostatic deflection)	Ib3 = 25 µAdc	Width	---	0.69	mm
	5246	Focusing voltage at modulation condition		Eb1	---	610	Vdc
Ⓑ	5101	Angle between traces		---	89	91	Degrees
Ⓑ	5106	Bulb, screen, and faceplate quality		---	---	---	---
Ⓑ	----	Trace alinement (gun to gun)	See note 7	---	---	---	---
Ⓑ	5216	Stray light emission (conventional types)	Eb2 = 3,500 Vdc; Eb3 = 10,500 Vdc	---	---	---	---
Ⓑ	5226	Line width A (electrostatic deflection)	Ib3 = 25 µAdc	Width	---	0.65	mm
Ⓑ	5246	Focusing voltage at cutoff		Eb1	---	610	Vdc
Ⓑ	5248	Deflection factor	1D2 (see note 8)	DF	54	66	Vdc/in.
Ⓑ	5248	Deflection factor	3D4 (see note 8)	DF	43	53	Vdc/in.
Ⓑ	5248	Deflection factor	1D2; Eb3 = Eb2 = 2,000 Vdc	DF	43	55	Vdc/in.
Ⓑ	5248	Deflection factor	3D4; Eb3 = Eb2 = 2,000 Vdc	DF	36	44	Vdc/in.
	5251	Heater-cathode leakage current		Ihk	---	15	µAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued					
1101	Secureness of base, cap, or insert		---	---	---	---
Ⓑ 1111	Base pin solder depth		---	---	---	---
1105	Permanence of marking		---	---	---	---
	Quality conformance inspection, part 3					
----	Life-test provisions	Group C; t = 500 hours (min); Ib3 = 25 μAdc; Eb2 = 3,500 Vdc; Eb3 = 10,500 Vdc	---	---	---	---
----	Life-test end points	Ib3 = 18.75 μAdc				
		Modulation	ΔEc	---	35	Vdc
		Line width A	Width	---	0.65	mm
		Line width B	Width	---	0.75	mm
		Grid No. 1 leakage	---	---	---	---
		Anode No. 2 leakage	---	---	---	---
		Heater-cathode leakage	---	---	---	---
		Anode No. 1 leakage	---	---	---	---
		Stray emission	---	---	---	---

NOTES:

1. Accelerator average power input shall be limited to 6 watts. Grid No. 4 and grid No. 2 are connected internally and referred to herein as accelerator. This shield and the accelerator electrodes should be connected together under normal operating and test conditions. However, the accelerator as well as the first ring of the post-deflection accelerating system may be operated at slightly different potentials for astigmatism control if desired and the shield potential should then be adjusted for optimum performance.
2. Deflection electrode circuit resistances should be equal. Higher resistance values up to 5.0 megohms may be used for low-beam current operation.
3. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity, pattern distortion, and tracking.
4. All tests except capacitance, vibration, tracking, cathode illumination, stray emission, and interaction factor shall be made on each unit separately. Each deflection plate shall be connected to A2 through a resistance of 1 megohm.
5. This test to be performed at the conclusion of the holding period.
6. Maximum potentials applied to base, deflection plates, and A2 only.
7. Corresponding traces of each unit shall be within 1° of each other.
8. The ratio of the deflection factors of corresponding deflection electrodes shall not exceed 1.15.

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NOTES: -Continued

9. The focused spots of both guns shall be moved by means of a common dc voltage among the 1D2 axis until the spot of gun A is at ±1 inch from the geometrical center of the tracking area. Then by means of a small correction voltage the spot of gun B shall be moved to the same distance. The spots then shall be moved by means of the common dc voltage in the opposite direction from the geometrical center until the spot of the less sensitive gun reaches -1 inch. The difference in deflection factor of the more sensitive gun is then compensated for by moving its spot back to -1 inch by means of a voltage divider. This entire procedure is to be duplicated along the 3D4 axis. Then by means of the two common dc voltages, the spots shall be tracked over the entire tracking area, a 4-inch diameter circle centered with respect to the tube face center, and the greatest distance between the spots shall not exceed the limit specified.

10. Centered with respect to the tube face and with the tube shielded.

11. With a raster pattern, adjusted so its widest points just touch the sides of a 3.060-inch square, will fall within the area bounded by the 3.060-inch square and an inscribed 2.940-inch square.

12. All other electrodes contained within the envelope are to be tied together.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 80
DSA - ES

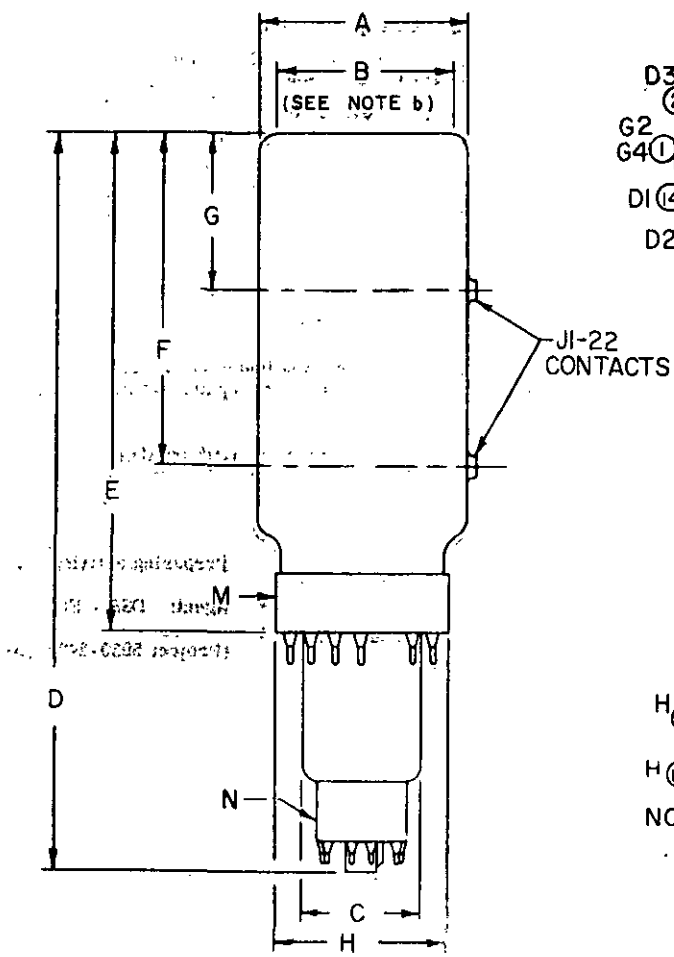
User activities:
Army - MU, WC
Navy - AS, OS, CG, SH
Air Force - 19

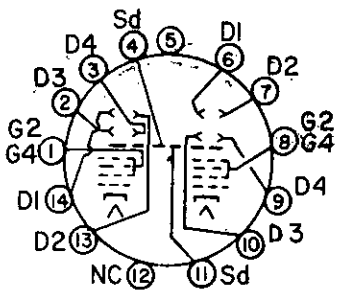
Preparing activity: Navy - EC

Agent: DSA - ES

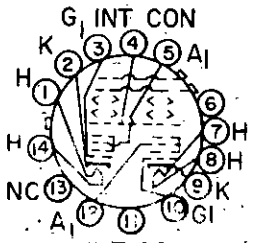
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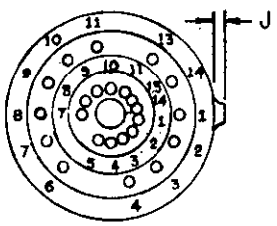


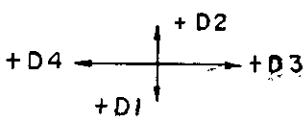
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DEFLECTION DIRECTIONS
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NOTES:

a. Shield (collar pins 4 and 11) is common to both beams.
b. Minimum useful screen diameter.

FIGURE 1. Outline drawing of electron tube types 5AFP1 and 5AFP7.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.16 (130.97)	5.37 (135.73)
B	4.50 (114.30)	
C	2.94 (74.61)	3.06 (77.79)
D	18.00 (457.20)	18.50 (469.90)
E	12.00 (304.80)	12.50 (317.50)
F	7.75 (196.85)	8.25 (209.55)
G	3.44 (87.31)	3.94 (100.01)
H	4.25 (107.95)	4.375 (111.13)
J		.125 (3.18)
Quality conformance inspection, part 3 (periodic check)		
K	Envelope: J42R1, J42S1	
L	Contact: J1-22	
M	Collarbase: A12-94	
N	Base: B12-37	

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